Koide's 2/3 as a Generation-Coherence Structure: an Event Density Re-description, a Toy Doublet-Grid Map, and a Neutrino Prediction

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Abstract

The Koide relation — that the three charged-lepton masses satisfy $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ to one part in 10^5 — is a genuine, complete-data regularity (there are exactly three generations; none is omitted). We record three things. First, a compact restatement: writing each fermion family in Brannen form places the three generations 120° apart on a circle, and the entire Koide phenomenology reduces to the one line $\mathbf{Q = 1/3 + w^2/6}$, where the "wobble depth" w is $\sqrt{2}$ for the charged leptons (the value that yields 2/3) and larger for the quark families. Second, an Event Density re-description: since mass is rest-frame bandwidth content (`Paper_113`) and ED's carrier is $P = \sqrt{b}\, e^{i\pi_K}$ (with π_K the polarity phase), the $\sqrt{m}$ in Koide's formula plays the role of the ED amplitude of a mass-mode — treating the three generations as coherent amplitude-channels, an organizing choice rather than a corpus result — so Q reads as a coherence ratio among three generation-amplitudes. This is a shape-match, not a derivation. Third, two toy maps that place the wobble over the fermions' real quantum numbers (electric charge; then weak isospin × hypercharge); both are underdetermined by the three charged data points, and both extrapolate to the same single prediction — a neutrino Koide value $\mathbf{Q \approx 0.77}$ — which is untestable until neutrino masses sharpen. We are explicit throughout that ED re-describes and toy-maps this structure; it does not derive it.

1. The relation, and the honest weight it carries

Koide (1981) observed that the three charged-lepton pole masses satisfy

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = 0.666659 \approx \tfrac{2}{3},$$

precise to ~1 part in 10^5, and *tightening* toward 2/3 as the τ mass has been remeasured over four decades. Two honest points frame everything below.

This is a complete-data regularity, not a small sample. There are exactly three charged-lepton generations; no fourth exists to measure, and none was left out. So Koide is the *entire* dataset producing one dimensionless number that lands on a simple fraction. That is why it is taken seriously and is not dismissible as a coincidence among many trials — there are no other trials.

It is clean only for the charged leptons. The up- and down-quark families give Q ≈ 0.85 and 0.73, but quark masses are scheme-dependent (running values), so those numbers are soft. Neutrino masses are essentially unknown. The rigid, remarkable fact is the charged-lepton 2/3.

2. The geometry, and one law behind every family

Write each family in the **Brannen (2005) form**:

$$\sqrt{m_k} = M\,(1 + w\cos(\delta + \tfrac{2\pi k}{3})), \qquad k = 0, 1, 2,$$

three points **120° apart on a circle** (a Z_3 / democratic arrangement) with overall scale M, wobble depth w, and phase offset δ. Each dot's horizontal projection is a $\sqrt{m}$ (Figure 1). Because the three cosines sum to zero and their squares sum to 3/2, one gets $\Sigma\sqrt{m} = 3M$ and $\Sigma m = M^2(3 + 3w^2/2)$, so the whole Koide quantity collapses to

$$\boxed{Q = \tfrac{1}{3} + \tfrac{w^2}{6}}.$$

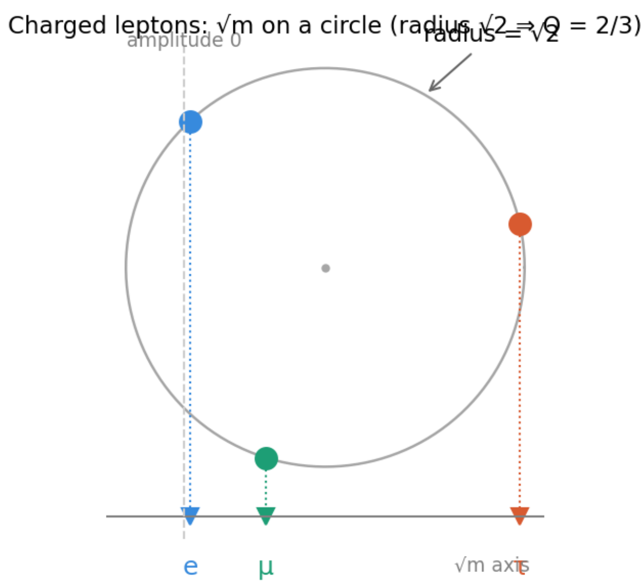

Figure 1: Figure 1. The three charged leptons as $\sqrt{m}$ on a circle of radius $\sqrt{2}$ (Brannen form). Each dot's horizontal position is a $\sqrt{m}$; the radius $\sqrt{2}$ is exactly the value that makes Q = 2/3, equivalently the $\sqrt{m}$ vector sitting at 45° to the democratic (1,1,1) direction (cos = $1/\sqrt{2}$).

So **Q depends only on the wobble depth w**, not on δ or M. The families differ only in w: charged leptons at $\mathbf{w} = \sqrt{2}$ (⇒ Q = 2/3 exactly — the $\sqrt{m}$ vector at 45° to the (1,1,1) diagonal, cos = $1/\sqrt{2}$), down-quarks at w ≈ 1.55, up-quarks at w ≈ 1.76 (Figure 2). The phase δ rotates the triangle and sets the *ratios* among the three masses (leptons: 1 : 207 : 3477, the real values) without touching Q. For the leptons $\delta \approx 0.222 \approx 2/9$ rad — a second, separate, unexplained near-coincidence, and **not** the source of the 2/3 (which holds for any δ).

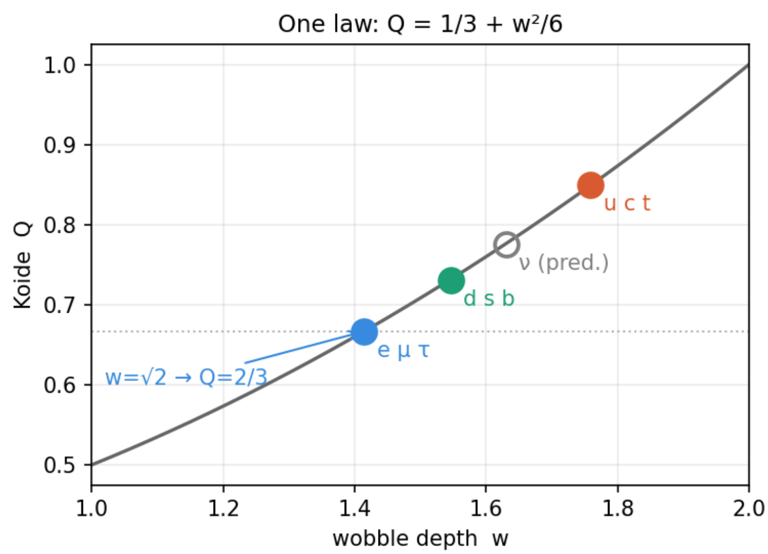

Figure 2: Figure 2. The single law Q = 1/3 + $w^2/6$. The charged leptons sit at w = $\sqrt{2}$ (Q = 2/3); the quark families at larger w; the open circle is the neutrino corner extrapolated from the toy maps of §4.

3. The Event Density re-description (a shape-match, not a derivation)

Koide's skeleton is (Σ m)/(Σ $\sqrt{m}$)^2 = (sum of squared amplitudes)/(coherent sum, squared). That is the **amplitude-vs-probability structure of quantum mechanics**: add amplitudes and square to get interference; sum the squares to get probabilities. Event Density is built on exactly this object. Its participation carrier is $P = \sqrt{b}\, e^{i\pi_K}$ (π_K the polarity phase), its Born rule reads $f = |P|^2$ (`Paper_003`), and mass is **rest-frame bandwidth content b** (`Paper_113`, tier M2). Hence $\sqrt{m}$ **maps onto the ED amplitude of a mass-carrying mode** — if the three generations are read as coherent amplitude-channels, an organizing assumption rather than a corpus result — and Koide's Q is then a **coherence / spread ratio among the three generation-amplitudes** — how democratically the three lepton modes share amplitude — with 2/3 a balanced value. Brannen's form then reads directly in ED terms: three amplitudes at 120° relative phase (Z_3), wobble depth $\sqrt{2}$, offset $\delta \approx 2/9$.

Tier. This is RE-DESCRIPTION. It re-expresses the *shape* of Koide (the roots, the square, the 1/3-to-1 range) in ED's own idiom, and nothing more; the Z_3 three-generation structure is an inherited *input* (Brannen's ansatz fitted to the three known generations), not something ED explains. It does not derive the masses, the generation count, w = $\sqrt{2}$, or $\delta \approx 2/9$. Its value is that it relocates the mystery into ED's native variables (amplitude, phase, coherence), which is where any future derivation would have to live — and coherence is the object ED's Σ functional and V5 budget already deal in.

4. Two toy maps over the real quantum numbers, and one prediction

Mass is not fundamental in the Standard Model; charge and weak isospin/hypercharge are, with mass downstream. So we ask whether the wobble w organizes over the *real* quantum numbers.

Charge (1-D). Against electric charge, the three charged families give (-1, 1.414), (-1/3, 1.546), (+2/3, 1.759): per unit charge the successive slopes are +0.20 and +0.21 — nearly equal, so **w is approximately linear in charge, w ≈ 1.61 + 0.20·Q_charge.** Put charge on a vertical axis and the wobble grows linearly with height — a straight-sided cone.

Isospin × hypercharge (2-D). More honestly, the fermions live on a 2×2 doublet grid: weak isospin $T_3 = \pm\frac{1}{2}$ across (up-member vs down-member), hypercharge Y up (quark doublet $+\frac{1}{3}$, lepton doublet -1). Placing a Koide triangle at each site (Figure 3), the wobble is **monotone in both directions** — up-members exceed down-members, quarks exceed leptons — and lies close to a plane over the grid, largest at the up-quark corner, smallest ($\sqrt{2}$) at the electron.

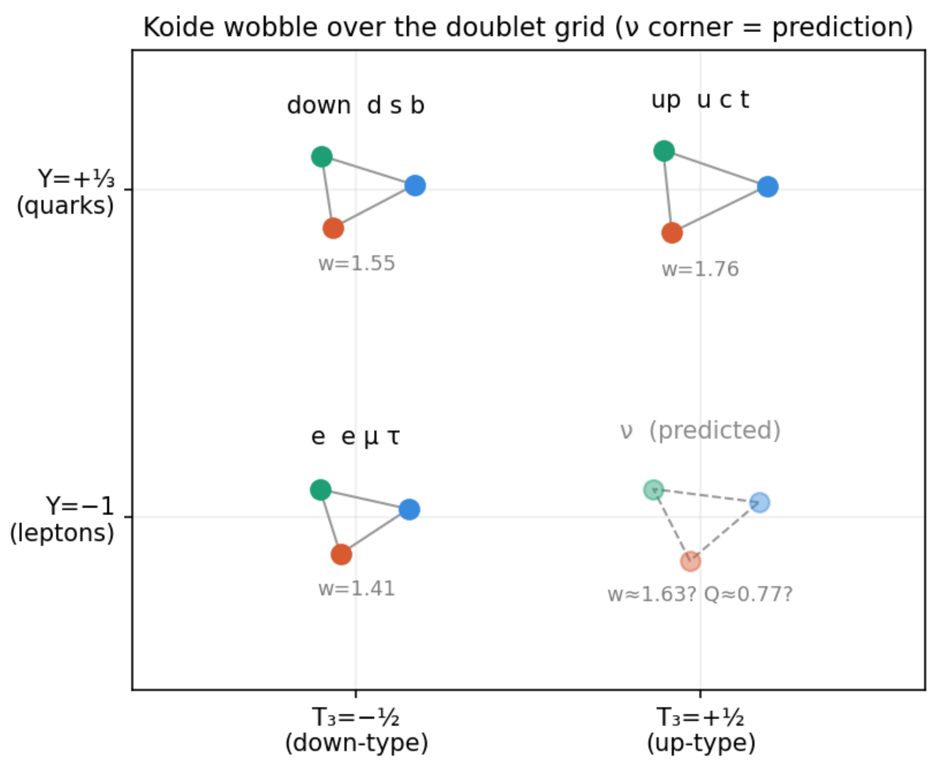

Figure 3: Figure 3. The four fermion types on their own Standard-Model coordinates (weak isospin across, hypercharge up), each a Koide triangle whose size is the wobble w. The wobble grows toward the up-quark corner. The dashed neutrino corner is a prediction, not data.

The one prediction. The plane through the three charged corners puts the neutrinos ($T_3 = +\frac{1}{2}$, Y = -1) at **w ≈ 1.63, Koide Q ≈ 0.77** — the dashed corner of Figure 3 and the open point of Figure 2. The 1-D charge line gives the same value. Neutrino masses are far too poorly known to test this today, but it is a concrete, falsifiable-in-principle output.

Tier — read this before believing the maps. These are TOY MODELS with soft legs.

The lepton 2/3 (§1–2) is rigid and complete-data; the cross-family maps are not. The plane has three free parameters and there are exactly three charged data points, so it fits *automatically* and carries no residual to test — its only content is the neutrino corner. Two of the three inputs (the quark w's) are scheme-fuzzy. And the Standard Model's structure is not truly 1-D: up and down quarks are isospin partners, so a single clean axis can only be partial. So Figure 3 is a legitimate, non-arbitrary *map* (it uses the SM's own labels, not invented ones) plus one filed prediction — not evidence of a law.

5. What connects to ED, and what does not

- **Compatible, at the level of structure.** ED grounds charge as integer topological winding (`Paper_ChargeAsTopology_B4`) and the SU(2)/doublet gauge structure from channel multiplicity, so the axes of Figure 3 are ED-adjacent, and every slice (a coherent superposition of three $\sqrt{m}$ amplitudes) is ED-native language.
- **Not derived.** ED does not produce the fractional hypercharges, the masses, the generation count, or the values $\mathrm{w} = \sqrt{2}$, $\delta \approx 2/9$. So the docking port — the map's axes and the reason the wobble takes the values it does — is exactly the part ED has not built.
- **The honest ED talking point (consilience-tier).** In the Standard Model, masses are arbitrary Yukawa dials, so Koide's 2/3 is a fluke among free parameters. In ED, mass *is* bandwidth content and $\sqrt{m}$ its amplitude, so "the lepton masses fold into a clean amplitude-on-a-circle relation" is the *kind* of thing expected if masses are amplitudes and a surprise if they are dials. That is a modest point of consilience for ED's ontology — not a derivation, and not to be inflated.

6. Bottom line

Read through Event Density, Koide's relation is a **$\mathbf{Z}_3$-symmetric, $\sqrt{2}$-deep coherence pattern among three generation-amplitudes**, governed entirely by $\mathrm{Q} = 1/3 + \mathrm{w}^2/6$, with one unexplained phase ($\delta \approx 2/9$). Over the fermions' real quantum numbers the wobble lays out as a smooth field — a legitimate map, with a single filed prediction (neutrino $\mathrm{Q} \approx 0.77$). ED can say this much: it re-describes the structure in its own amplitude/coherence language and offers an honest toy map. It cannot derive the value, the generation count, or any mass — those are the Standard Model flavor sector, which ED inherits. The two things worth remembering are the **complete-data weight of the lepton 2/3** and the **neutrino prediction**, either of which becomes actionable only when neutrino masses sharpen or when ED's gauge-and-generation sector is derived rather than inherited.